

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
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**THE KIOSK GUYS
SMART KIOSK**

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CONTENTS

1 Problem Statement	6
2 Methodology	6
3 Value Proposition	6
4 Development Milestones	7
5 Background	7
6 Related Work	8
7 System Overview	9
8 Roles & Responsibilities	10
9 Cost Proposal	10
9.1 Preliminary Budget	10
9.2 Current & Pending Support	11
10 Facilities & Equipment	11
11 Assumptions	11
12 Constraints	11
13 Risks	11
14 Documentation & Reporting	12
14.1 Major Documentation Deliverables	12
14.1.1 Project Charter	12
14.1.2 System Requirements Specification	12
14.1.3 Architectural Design Specification	12
14.1.4 Detailed Design Specification	12
14.2 Recurring Sprint Items	12
14.2.1 Product Backlog	12
14.2.2 Sprint Planning	13
14.2.3 Sprint Goal	13
14.2.4 Sprint Backlog	13
14.2.5 Task Breakdown	13
14.2.6 Sprint Burn Down Charts	13
14.2.7 Sprint Retrospective	13
14.2.8 Individual Status Reports	13
14.2.9 Engineering Notebooks	13
14.3 Closeout Materials	13
14.3.1 System Prototype	13
14.3.2 Project Poster	14
14.3.3 Web Page	14

14.3.4 Demo Video	14
14.3.5 Source Code	14
14.3.6 Source Code Documentation	15
14.3.7 Hardware Schematics	15
14.3.8 CAD files	15
14.3.9 Installation Scripts	15
14.3.10Admin Operation Instructions	15

LIST OF FIGURES

1	Sprint 1 Burndown Chart	14
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1 PROBLEM STATEMENT

The UTA Engineering Research South (ERSA) building currently lacks a centralized and convenient system for managing conference room reservations, office supply requests, and staff/room directory services. This can lead to scheduling conflicts, inefficiencies in ordering and tracking supplies, and difficulties for visitors in locating rooms or offices. By addressing these gaps, the ERSA building staff can improve productivity, reduce wasted time, and provide a more user-friendly experience for faculty, staff, and visitors.

Additionally, the UTA Engineering Research Building (ERB) currently lacks directory services for guests trying to reach their desired destination. This problem could lead to negative guest experiences, such as being late to meetings and feelings of displeasure. By providing a solution, faculty and visitors can have a greater experience by reducing time and having an overview of the building.

2 METHODOLOGY

Our team will design and develop a smart self-service kiosk for the UTA Engineering Research South (ERSA) building to streamline key building services. The kiosk will provide three main functions:

1. **Interactive Directory and Navigation** – A searchable staff and room directory that displays the building's map with highlighted paths to help visitors quickly locate offices and conference rooms.
2. **Conference Room Scheduling System** – An interactive interface that allows users to view room availability, make on-the-spot reservations, and reduce scheduling conflicts by syncing with existing calendar systems.
3. **Supply Request Feature** – A one-tap menu where staff can request office supplies, with automatic notifications sent to the supply manager.

The kiosk will serve as a central access point for faculty, staff, and visitors, increasing efficiency, reducing wasted time, and improving the overall user experience within ERSA. This kiosk will also be designed with the stakeholders in mind to ensure all accessibility standards are met.

To accompany the interactive smart kiosk for the ERSA building, there will be an online remote version, where users are able to manage room reservations, and administrators will be able to access admin privileges, such as editing supplies, user profiles, and kiosk banners.

Lastly, our team will also design and develop two smart interactive kiosks for the UTA Engineering Research Building (ERB), where they will be located on the West and East sides, respectively. The two kiosks for the ERB will simply provide an interactive directory and way-finding feature, similar to the ERSA kiosk, for users to navigate around the venue. Similarly, this kiosk will be designed with the stakeholders in mind to ensure all accessibility standards are met.

3 VALUE PROPOSITION

The Smart Kiosk systems will provide significant value to both the ERSA and ERB, and their stakeholders by improving efficiency, communication, and user experience. By combining conference room scheduling, supply management, and directory services into a centralized platform, the kiosk will streamline everyday tasks, minimizing administrative overhead and time wasted.

For faculty and staff, the ERSA kiosk simplifies scheduling and resource requests, ensuring that meetings and supply needs are met without conflict or delay. For building managers and administrators, the

system offers data-driven insights on room usage and supply consumption, allowing proactive decision-making and cost savings. The way-finding features of both ERSA and ERB kiosks provide quick navigation and a more welcoming experience to their respective locations.

Overall, these smart kiosks will improve daily operations, satisfaction for all users, and accentuate UTA's commitment to innovation and efficiency in campus services.

4 DEVELOPMENT MILESTONES

- Beginning of Sprint 1 - 09 September 2025
- Project Charter First Draft - 26 September 2025
- Beginning of Sprint 2 - 29 September 2025
- System Requirements Specification First Draft - 17 October 2025
- Beginning of Sprint 3 - 20 October 2025
- Architectural Design Specification First Draft - 07 November 2025
- Beginning of Sprint 4 - 10 November 2025
- Beginning of Sprint 5 - 26 January 2026
- Detailed Design Specification First Draft - 13 February 2026
- Beginning of Sprint 6 - 16 February 2026
- Beginning of Sprint 7 - 16 March 2026
- Beginning of Sprint 8 - 06 April 2026
- Innovation Day - 14 April 2026
- Demo Day - 24 April 2026

5 BACKGROUND

The University of Texas at Arlington's Engineering Research South (ERSA) building is a space for research, collaboration, and administrative activity. However, due to its recent construction, the building currently lacks a convenient centralized system to manage daily needs such as reserving conference rooms, requesting office supplies, and locating staff offices. These inefficiencies affect the productivity of the staff.

Currently, room reservations are handled in a fragmented manner, often requiring users to rely on email exchanges, shared calendars, or direct communication with administrators. This approach creates opportunities for scheduling conflicts, double-bookings, and miscommunication. As the ERSA building has multiple staff members who schedule meetings, the lack of a reliable scheduling system wastes valuable time and reduces productivity. In addition, office supply management is performed manually, in which staff have to manually request supplies themselves. This often results in delays, uncertainty about inventory levels, and unnecessary back-and-forth communication between staff and supply managers. Without an efficient way to track requests or monitor usage trends, supply shortages can occur

unexpectedly, interrupting workflows. Potential new members as well as existing members of ERSA face yet another challenge: locating an unfamiliar conference room. With no path specified, individuals may struggle to locate specific offices or conference rooms, creating frustration and inefficiency. A solution that provides clear directions and room information would improve the user experience and reduce interruptions for staff who are frequently asked for directions.

The motivation for this project comes directly from the needs of ERSA faculty, staff, and building administrators, who have expressed interest in a centralized system that streamlines these processes. By developing a smart kiosk, our team has the opportunity to provide a solution that addresses these problems while also showcasing UTA's Computer Science and Engineering department's commitment to innovation and campus initiatives. This project is supported by Dr. Wang, who connected our team with our project sponsors, Jade Myers and J.D Briggs, to ensure alignment with real operational needs. Our role as a development team is to bridge this gap between inefficiency and innovation by providing a practical and user-friendly system. The kiosk project is not only an academic exercise in software and hardware integration, but also a tangible step toward improving daily operations and overall user experience in ERSA.

6 RELATED WORK

While brainstorming potential avenues and ideas for what we would like our end product to look like and operate, we discovered many real-world examples to evaluate and draw inspiration from.

The first example that immediately came to our mind was the self-service kiosks McDonald's has deployed all across the world. These kiosks are designed to enhance the customer's experience through their intuitive interface, which leads to improved wait times, eliminating interactions with workers, and providing more details in regards to their products. [1] While McDonald's kiosks are widely popular, this solution would not be practical due to our product due to their large size and budget constraints. McDonald's kiosks are rather large and if we were to implement a design similar to it, it might appear too out of place at the ERSA building and the materials used to create it would be too complex and expensive for the desired functionalities.

The next real-world solution that we evaluated is the reservation kiosks local to us at the University of Texas at Arlington. These kiosks are at the School of Social Work and Smart Hospital (SWSH) building, The Commons, and each of the libraries. These kiosks are specialized to allow students and staff to reserve study spaces and display the current status of the respective rooms. However, due to their specialized functionality and small size, they are often overlooked and even perceived as a widget to some users. Furthermore, the smaller size would not work for our stakeholders, as trying to solve their needs would result in an aesthetically displeasing product to use, leading to users steering away from the product. Another reason why these designs would not be valid for our stakeholders would be due to the way the kiosk touchscreen is mounted. This kiosk differs from the other solution mentioned earlier as it is mounted on the wall, which is against our stakeholders' preferences.

Wayfinding kiosks provide users with an interactive map of their surrounding area and, in many cases, the most direct path from point A to point B. These kiosks have been deployed in a variety of public spaces such as airports, malls, and schools. For example, Meridian, a company specializing in kiosk solutions, implemented its wayfinding kiosk at the Port Columbus International Airport to help visitors

navigate both the airport and the city. [2] Our product will adopt a similar approach by offering an interactive map and navigation system specifically for the UTA ERSA building. However, unlike Meridian's wayfinder kiosk, which is limited to map-based navigation, our system will go beyond basic wayfinding. It will also include additional features such as supply tracking and a conference room reservation system, creating a more versatile and multifunctional solution tailored to the needs of the building's users.

One of the challenges that we have identified for our product is the implementation of the interactive directory and navigation function that was provided to us by our product owner. In our research for indoor navigation and mapping we found an article that highlights the devices and software applications used for accessible wayfinding and navigation. While the article delves into detail regarding navigation and wayfinding, what was discovered is that in terms of navigation/wayfinding, typically only disabled people were considered in these studies rather than the general population. [3] Thus, these methods primarily focused on aiding those with disabilities.

7 SYSTEM OVERVIEW

To address the current challenges in the ERSA building, our team will develop a web-based smart kiosk system that integrates three primary functions: conference room scheduling, supply requests, and interactive directory/navigation. The system will be accessible through both a physical kiosk located in ERSA and via personal devices such as laptops and smartphones.

1. **User Interface (Kiosk Website)** – A touchscreen-friendly web application will serve as the central access point. By default, users will see an interactive building map with room locations. From here, they can either navigate the directory or log in to access additional features.
2. **Authentication and User Accounts** – To ensure security and proper access, the system will require users to create an account linked to a valid institutional domain (e.g., uta.edu). Accounts must be confirmed through an email verification process. Once registered, users will have access to the full functionality of the kiosk. A password reset option will also be available through email. Users are able to connect their accounts to their UTA ID, to have the option to quickly log in by swiping their card.
3. **Conference Room Scheduling** – Logged-in users can view availability and schedule conference rooms. The scheduling data will be stored in a central database, helping avoid double-bookings and ensuring accurate availability.
4. **Supply Request System** – Using a simple request form, users can request office supplies. Each request will automatically generate an email notification to the supply manager, including the requesting user's information and the room associated with the request. Usage data can also be tracked within the database to help predict supply restocking needs.
5. **Idle Banner Slideshow** – When the smart-kiosks for both ERSA and ERB are in idle mode, they will display a rotating slideshow of banners that promote UTA, UTA's CSE department, and other campus events.
6. **Database and Backend Services** – A cloud database will store user accounts, room schedules, supply request data, and images. This enables consistent access whether users are at the kiosk or using the website remotely.
7. **Multi-Platform Access** – While the kiosk serves as the primary access point, the system will also be available remotely, allowing faculty and staff to use its features from their own devices when away from the kiosk.

8 ROLES & RESPONSIBILITIES

The initial stakeholders of this project are the UTA Computer Science and Engineering staff and those who have access to the Engineering Research South A (ERSA) building. The sponsor behind this project is Jade Myers and J.D Griggs, two of the main assistants of the computer science department. All questions regarding the scope of the product will be directed towards them, and the other stakeholders in mind. The project was further expanded to include the two smart kiosk systems for the UTA ERB, which resulted in the stakeholders growing to include the faculty, students, and other visitors of the building. The main sponsors of our project remained Jade Myers and J.D Griggs as we reported all progress and communication with them to ensure a stakeholder requirements were met.

The team in charge of designing and constructing this project consists of five undergraduate computer science students at the University of Texas at Arlington: Miguel Cervantes, Ivan Ko, Thien Nguyen, Earl Sanchez, and Prakash Sapkota. Since this project is sponsored by UTA's CSE department, the sponsor is the designated product owner. Our team recognized the importance of appointing a scrum master to ensure that scrum practices are being implemented and to appropriately manage workflow. This resulted in Miguel Cervantes being unanimously voted as the best fit for the position. We all agreed to act diligently to maintain these roles for the duration of the project.

9 COST PROPOSAL

To determine the overall cost of the project, a thorough amount of research must be completed, as money is a vital and scarce resource among our team and sponsors. It is important to us that we do not waste this resource on unnecessary items or services throughout the duration of this project. Based on what we initially researched and evaluated, we initially approximated that our cost for each kiosk would range somewhere between **\$640 – \$1050**. We came to this conclusion based on the current market value for the products we believe would be suited best for the product, along with what we have agreed to individually allocate towards it. We are being allocated a budget of \$800.00 for the kiosk by the Senior Design department, as well as being funded by our sponsors at an undisclosed number. The cost of the products we have been interested in is listed below.

This budget is tentative and subject to change at any moment.

9.1 PRELIMINARY BUDGET

Item	Quantity	Estimated Cost
Touchscreen Monitor	3	\$250 – \$350
Single-Board Computer	3	\$150 – \$250
Kiosk Stand	3	\$200 – \$350
Miscellaneous Cables/Adapters	–	\$30 – \$50
Application Subscriptions	–	\$10 – \$50
Total Estimated Cost		\$640 – \$1050

Table 1: Preliminary budget for smart kiosk components

9.2 CURRENT & PENDING SUPPORT

The CSE department is providing \$800.00 in funding to our project. Our current budget is centered around this allotted amount for the construction and integration of the smart kiosk. After the introduction of the two ERB kiosks, our project's funding has been boosted by an undisclosed amount in order for the group to successfully reach stakeholder requirements.

10 FACILITIES & EQUIPMENT

Throughout the duration of the project, our group utilized the labs in the ERB building to hold meetings, complete project work, assemble, and ultimately bring the project to fruition. At the start, ERB 335A was utilized the most, but as the project progressed, we transitioned into lab ERB 203 to further accommodate our research and meeting needs.

11 ASSUMPTIONS

The following list contains critical assumptions related to the implementation and testing of the project.

- The team will be able to utilize a Raspberry Pi or similar single-board computer with built-in Wi-Fi/Ethernet, allowing straightforward network connectivity.
- The network infrastructure of both the ERSA and ERB buildings will provide reliable Wi-Fi/Ethernet access to the kiosk, without requiring special exceptions.
- Only faculty and staff will have access to the ERSA kiosk and users will adhere to professional standards when making reservations (avoiding misuse, such as reserving all rooms).
- Users of the ERB kiosk will adhere to professional standards when accessing the smart device.
- The supply manager will be emailed directly whenever a user requests items for a room.
- Adequate physical space and power outlets will be available in both the ERSA and ERB buildings for kiosk integration.

12 CONSTRAINTS

The following list contains key constraints related to the implementation and testing of the project.

- A final prototype demonstration must be completed by May 2026 (end of the Spring 2026 semester).
- The project must meet all requirements and standards set by the sponsor.
- Hardware and software resources will be limited to what is accessible through the team's budget, university resources, or free/open-source tools.
- Items may get delayed or lost in transit.
- Development and testing within the ERSA building must be scheduled with our sponsors, due to the building's swipe card access.

13 RISKS

The following high-level risk census contains identified project risks with the highest exposure. Mitigation strategies will be discussed in future planning sessions.

Risk Description	Probability	Loss (days)	Exposure (days)
Availability of hardware parts due to vendor delay	0.40	30	12
Sponsor unavailable for timely feedback or clarification	0.10	5	0.5
Network access (Wi-Fi/Ethernet) not reliable in ERB and ERSA	0.20	7	1.4
Team member unavailable due to illness or scheduling conflict	0.25	2	0.5
Integration issues between scheduling system and calendar APIs	0.10	15	1.5

Table 2: Overview of highest exposure project risks

14 DOCUMENTATION & REPORTING

14.1 MAJOR DOCUMENTATION DELIVERABLES

14.1.1 PROJECT CHARTER

The Project Charter will be maintained through Overleaf, a cloud-based editor for LaTeX documents, and will be updated periodically when major changes regarding the scope of the project are made. The initial version will be delivered on the final day of the first sprint, September 26th, 2025. The final version will be delivered as part of the final deliverables by April 24th, 2026.

14.1.2 SYSTEM REQUIREMENTS SPECIFICATION

The System Requirements Specification (SRS) will be initiated at the start of sprint 2 and will be the main focus during that period. The document will be created and maintained through Overleaf. The initial version will be delivered by the end of the second sprint, October 17th, 2025. This document will be reviewed periodically, and any changes to the system's functional and non-functional requirements will be updated in the document as needed. The final version will be delivered as part of the final deliverables by April 24th, 2026.

14.1.3 ARCHITECTURAL DESIGN SPECIFICATION

The Architectural Design Specification (ADS) will be initiated at the start of sprint 3, and will be the major priority during that period. The document will be created and maintained through Overleaf. The initial version will be delivered at the end of the third sprint, November 7th, 2025. This document will be reviewed periodically, and any changes to the high-level system architecture or component interactions will be updated in the document as needed. The final version will be delivered as part of the final deliverables by April 24th, 2026.

14.1.4 DETAILED DESIGN SPECIFICATION

The Detailed Design Specification (DDS) will be initiated at the start of sprint 4, and will be the main focus of that period. The DDS will provide an extensive breakdown of the system components and implementation methods. Because there is currently no mention of the document in the course schedule, the initial version will be subject to delivery in early February 2026. This document will be reviewed periodically, and any changes to the system components and implementation methods will be updated in the document as needed. The final version will be delivered as part of the final deliverables by April 24th, 2026.

14.2 RECURRING SPRINT ITEMS

14.2.1 PRODUCT BACKLOG

Items specified in the System Requirements Specification (SRS) will be implemented into the product backlog. These items will be prioritized based on factors such as complexity, estimated effort, and overall importance to the project. This will be decided collectively as a team with confirmation from

the product owner. The product backlog will be documented and shared using collaborative tools such as Microsoft Excel and Google Docs, allowing all team members and stakeholders to review updates.

14.2.2 SPRINT PLANNING

This project will be broken down into eight total sprints. Senior Design 1 consists of four sprints, and each sprint will be planned on the Friday before it commences. These sprint planning sessions will occur in person in the Engineering Research Building (ERB) 335A. Senior Design 2 consists of the other four sprints and will follow a similar planning process to the one implemented in Senior Design 1.

14.2.3 SPRINT GOAL

The sprint goal will be decided collectively by the team during the sprint planning sessions and finalized by our scrum master, Miguel Cervantes. Any concerns with the sponsor and other stakeholders will be clarified when deciding the goal.

14.2.4 SPRINT BACKLOG

The team will decide collectively which product backlog items make their way into the sprint backlog during the sprint planning sessions. The backlog will be maintained through a collaborative Google Sheets spreadsheet and Google Slides PowerPoint to gather a strong understanding of completion progress.

14.2.5 TASK BREAKDOWN

Individual tasks will be discussed and assigned from the sprint backlog on a voluntary or first-come, first-served basis. Each person's skills will be considered when assigning tasks. Documentation of time spent on these tasks will be maintained through a collaborative Google Sheets spreadsheet and Google Slides PowerPoint.

14.2.6 SPRINT BURN DOWN CHARTS

The Sprint Burn Down Charts for each sprint will be generated and managed by Ivan Ko. Ivan is able to access the data needed for the chart through our collaborative Google Sheets spreadsheet, where each team member will report the total amount of effort spent on tasks. A line graph will be used to display this data effectively.

14.2.7 SPRINT RETROSPECTIVE

Sprint retrospectives are dubbed as sprint reviews for this project. Sprint reviews are conducted in the final week of each sprint through a presentation in which our team reviews what we achieved this sprint. Additionally, each team member submits an individual sprint report, detailing what they accomplished during the sprint lifetime.

14.2.8 INDIVIDUAL STATUS REPORTS

Individual status reports will consist of individual sprint reviews as well as a peer review. Individual sprint reviews are reported at the end of each sprint cycle and will be conducted through a simple Microsoft Word document.

14.2.9 ENGINEERING NOTEBOOKS

Our team does not utilize any engineering notebooks for the project.

14.3 CLOSEOUT MATERIALS

14.3.1 SYSTEM PROTOTYPE

The final system prototype will include the final product as well as its deliverables. It will be demonstrated during Senior Design Demo Day in April 24th, 2026.

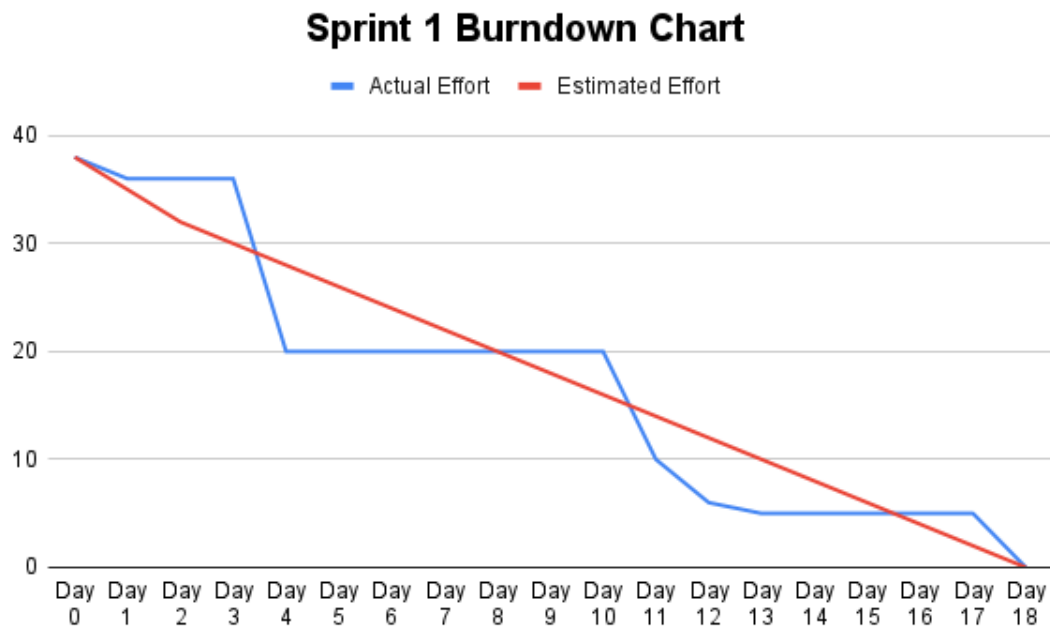


Figure 1: Sprint 1 Burndown Chart

14.3.2 PROJECT POSTER

A large-format project poster will be created to summarize the overall purpose, functionality, and user interaction of the smart-kiosks. The poster will include an overview of the project, the problem statement, methodology, results, system features, and high-level diagrams illustrating how users interact with the kiosk. Visuals such as screenshots, mock-ups, and schematics will also be included to make the poster more engaging.

The final poster will follow a landscape orientation that is curated through a Google Slides PowerPoint. The completed poster will be delivered and displayed on Innovation Day as well as Demo Day in April 2026, alongside prototypes of the Smart Kiosks demonstration.

14.3.3 WEB PAGE

The project web page will contain the project's overview as well as the project's final deliverables. Our team's information will be posted on the UTA Senior Design knowledge base by Steven McDermott.

14.3.4 DEMO VIDEO

Our project will contain a video demonstration of our products. Our team will provide an overview of what each respective kiosk's functionality entails and how to properly utilize the smart devices. This will include the core functionalities of user registration, login and verification, way-finding, room reservations, and the supply request system.

14.3.5 SOURCE CODE

All source code for the smart kiosks will be maintained and managed on GitHub. This repository will enable proper version tracking, collaborative development, and rollback capabilities if needed. We plan to set up automated deployment through Platform as a Service (PaaS) platforms so that any approved

changes to the repository will be reflected on the hosted kiosk website in real time.

Our project sponsor will be provided access to the repository, ensuring transparency and allowing them to review, download, or manage the code base as needed. At the conclusion of the project, the full repository will be turned over to the sponsor.

Licensing terms will be determined in consultation with the sponsor. If the project is released to the general public, an open-source license such as MIT or GPL may be applied. License details will be documented in a central README.md file, with references in each source file header where appropriate.

14.3.6 SOURCE CODE DOCUMENTATION

Source code documentation will be located in a README.md file describing how to execute the program. These generally use Markdown for formatting text. Additionally, comments will be included in the code to provide readability and context for the reader.

14.3.7 HARDWARE SCHEMATICS

Our project will not require the design or fabrication of custom printed circuit boards (PCBs). Instead, we plan to use standard hardware components, specifically a Raspberry Pi (or a similar single-board computer) connected to a multi-touch touchscreen monitor. These will be connected through standard cables (HDMI, power supply, and USB-C, USB-A) to run the web-based interface of the kiosk.

As such, no detailed PCB schematics will be necessary. A simple wiring diagram may be prepared to illustrate how the Raspberry Pi, touchscreen, and peripheral devices (e.g., power supply, network connection) are connected, but the hardware design itself will rely on commercially available parts.

14.3.8 CAD FILES

No CAD files were used during the duration of our project.

14.3.9 INSTALLATION SCRIPTS

The team will configure and provide a start-up script to load the program automatically onto the Raspberry Pi computers. The scripts will be kept alongside the application's source code on the GitHub repositories.

14.3.10 ADMIN OPERATION INSTRUCTIONS

Upon project completion, we will provide the product owners with an administrator operation instruction set that details the aspects of the smart-kiosks and a full-explanation of how to run the system to its full capacity.

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